

AM

intelligence is not far behind? No, it just means that chess was over-rated as an intellectual exercise! On a scale of 1 to 10, in terms of intellectual effort involved in playing the game, chess seems to rate at about .002. In terms of skill, concentration ability, depth of understanding of the game, etc. it is difficult. But then, so is multiplying two 37 digit numbers in your head difficult. Unless you're an "idiot savant", or a computer!

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Subject: Lenat's AM program

The exact reasons for the success of AM (and for its eventual failure to continue making new discoveries) have not been established. In Lenat's dissertation, he speculated that the source of power was the search heuristics, and that the eventual failure was caused by the inability of the system to generate new heuristics.

Then, in a paper by Lenat and Brown, a different reason is given: namely that the representation of concepts was the critical factor. There is a close relationship between mathematics concepts and lisp, so that mathematical concepts can be represented very concisely as lisp functions. Simple syntactic mutation operations, when applied to these concise functions, yield other interesting mathematical concepts. In new domains, such as those tackled by Eurisko, it was necessary to engineer the concept representation so that the concepts were concisely representable.

Finally, in a paper published this year by Lenat and Feigenbaum, yet another explanation of AM's (and Eurisko's) success and failure is given: "The ultimate limitation was not what we expected (cpu time), or hoped for (the need to learn new representations), but rather something at once surprising and daunting: the need to have a massive fraction of consensus reality already in the machine."

The problem with all of these explanations is that they have not been subjected to rigorous experimental and analytical tests, so at the present time, we still (more than ten years after AM) do not understand why AM worked!

I have my own pet hypothesis, which I am currently subjecting to an experimental test. The hypothesis is this: AM succeeded because its concept-creation operators generated a space that was dense in interesting mathematical concepts. This hypothesis contradicts each of the preceding ones. It claims that heuristics are not important (i.e., a brute force search using the concept-creation operators would be only polynomially--not exponentially--more expensive). It claims that the internal representation of the concepts (as lisp functions) was also unimportant (i.e., any other representation would work as well, because mutation operators are very rarely used by AM). Finally, it claims that additional world knowledge is irrelevant (because it succeeds without such knowledge).

There is already some evidence in favor of this hypothesis. At CMU, a student named Weimin Shen has developed a set of operators that can rediscover many of AM's concepts. The operators are applied in brute force fashion and they discover addition, doubling, halving, subtraction, multiplication, squaring, square roots, exponentiation, division, logarithms, and iterated exponentiation. All of these are discovered without manipulating the internal representation of the starting concepts.

AM is a "success" of AI in the sense that interesting and novel behavior was exhibited. However, it is a methodological failure of AI, because follow up studies were not conducted to understand causes of the successes and failures of AM. AM is not unique in this regard. Follow-up experimentation and analysis is critical to consolidating our successes and extracting lessons for future research. Let's get to work!

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